Course Code : ECT 355-3 **UVXV/MW – 24 / 1589**

Fifth Semester B. Tech. (Electronics and Communication Engineering) Examination

WIRELESS COMMUNICATION

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) Assume suitable data wherever necessary.

(2) Illustrate your answers with neat sketches, wherever necessary.

(3) Use of Erlang B Chart is permitted.

(4) All questions carry marks as indicated against them.

1. (a) Discuss handoff strategy in detail. 5(CO1)

(b) A city with coverage of 500 sq·km is covered with 12 cell system each with a radius of 1·387 km. If the total spectrum allocated is 28·5 MHz, with a full duplex channel bandwidth 25 KHz. Assume GOS of 2% for Erlang B system is specified and offered traffic per user is 0·03 Erlangs.

Compute :

(i) No. of cells in the service area.

(ii) No. of channels per cell.

(iii) Traffic intensity per cell.

(iv) Maximum carried traffic.

(v) Total no. of users that can be served for 2% GOS.

5(CO1)

2. (a) Deduce the equation for 2 Ray Propagation model. 5(CO2)

(b) If a WLAN transmitter operating at 2·4 GHz is located at the ground floor of ECE department in our campus and the power transmitted by the transmitter is 10 dBm, what is the received power in dBm at a distance of 10, 20, 30 meters away from the transmitter. Assume unity gain antennas and consider free space propagation. 5(CO2)

3. (a) Illustrate the time, frequency and Space diversity in brief. 5(CO3)

(b) Find 3-dB bandwidth for Guassian low pass filter used to produce 0·25 GMSK with a channel data rate of Rb = 270 kbps. What is the 90% power bandwidth in the RF channel ? 5(CO3)

4. (a) Compare wide band and narrow band communication systems in detail. 4(CO3)

(b) Consider a global system for Mobile, which is TDMA / FDD system that uses 25 MHz forward link, which is broken into radio channels of 200 KHz. If 8 Speech channels are supported on a single radio channel and if no guard band is assumed, find the number of simultaneous user that can be accommodated in GSM ? 3(CO4,5)

(c) In US AMPS, 416 channels are allocated to various cellular operators. The channels between them are 30 KHz with the guard band of 10 KHz. Calculate the spectrum allocation given to each operator. 3(CO4)

5. (a) A normal GSM has 3 start bits, 3 stop bits, 26 training bits for allowing adaptive equalization, 8·25 guard bits and 3 burst of 58 bits of encrypted data which is transmitted at 270·833 kbps in the channel.

Find :

(i) Number of overhead bits per frame.

(ii) Total Number of bits per frame.

(iii) Frame rate.

(iv) Time duration of a slot.

(v) Frame efficiency. 5(CO4,5)

(b) Explain the sequence of steps performed in GSM architecture to establish a call from one mobile user to another mobile user that belongs to different MSC. Use the block schematic of GSM architecture to explain above steps. 5(CO4,5)

6. Demonstrate the elegance of multicarrier modulation in OFDM for Delay Spread, Inter symbol Interference. 10(CO5)

Erlang B Traffic Table: *Maximum offered load versus B and S*

S/B	0.01%	0.05%	0.1%	0.5%	1%	2%	5%	10%	15%	20%	30%	40%
1	.0001	.0005	.0010	.0050	.0101	.0204	.0526	.1111	.1765	.2499	.4285	.6666
2	.0142	.0321	.0457	.1053	.1526	.2234	.3813	.5954	.7962	.9999	1.449	2.000
3	.0868	.1517	.1938	.3490	.4554	.6022	.8994	1.271	1.602	1.930	2.633	3.480
4	.2347	.3623	.4393	.7012	.8694	1.092	1.525	2.045	2.501	2.945	3.890	5.021
5	.4519	.6486	.7621	1.132	1.361	1.657	2.218	2.881	3.454	4.010	5.188	6.595
6	.7282	.9956	1.146	1.622	1.909	2.276	2.960	3.758	4.444	5.108	6.513	8.190
7	1.054	1.392	1.578	2.157	2.501	2.935	3.738	4.666	5.461	6.230	7.856	9.799
8	1.422	1.830	2.051	2.730	3.127	3.627	4.543	5.597	6.498	7.369	9.212	11.42
9	1.826	2.302	2.557	3.333	3.782	4.345	5.370	6.546	7.551	8.521	10.58	13.04
10	2.260	2.803	3.092	3.960	4.461	5.084	6.215	7.510	8.616	9.685	11.95	14.68
11	2.721	3.329	3.651	4.610	5.160	5.841	7.076	8.487	9.691	10.86	13.33	16.31
12	3.207	3.878	4.231	5.279	5.876	6.614	7.950	9.474	10.78	12.04	14.72	17.95
13	3.713	4.446	4.830	5.963	6.607	7.401	8.834	10.47	11.87	13.22	16.11	19.60
14	4.239	5.032	5.446	6.663	7.352	8.200	9.729	11.47	12.96	14.41	17.50	21.24
15	4.781	5.634	6.077	7.375	8.108	9.009	10.63	12.48	14.07	15.61	18.90	22.89
16	5.339	6.250	6.721	8.099	8.875	9.828	11.54	13.50	15.18	16.81	20.30	24.54
17	5.911	6.878	7.378	8.833	9.651	10.66	12.46	14.52	16.29	18.01	21.70	26.19
18	6.496	7.519	8.045	9.578	10.44	11.49	13.38	15.55	17.40	19.21	23.10	27.84
19	7.092	8.169	8.724	10.33	11.23	12.33	14.31	16.58	18.52	20.42	24.51	29.50
20	7.700	8.831	9.411	11.09	12.03	13.18	15.25	17.61	19.65	21.63	25.92	31.15
21	8.318	9.501	10.11	11.86	12.84	14.04	16.19	18.65	20.77	22.85	27.32	32.81
22	8.946	10.18	10.81	12.63	13.65	14.90	17.13	19.69	21.90	24.06	28.73	34.46
23	9.583	10.87	11.52	13.42	14.47	15.76	18.08	20.74	23.03	25.28	30.14	36.12
24	10.23	11.56	12.24	14.20	15.29	16.63	19.03	21.78	24.16	26.50	31.56	37.78
25	10.88	12.26	12.97	15.00	16.12	17.50	19.98	22.83	25.30	27.72	32.97	39.43
26	11.54	12.97	13.70	15.79	16.96	18.38	20.94	23.88	26.43	28.94	34.38	41.09
27	12.21	13.69	14.44	16.60	17.80	19.26	21.90	24.94	27.57	30.16	35.80	42.75
28	12.88	14.41	15.18	17.41	18.64	20.15	22.87	25.99	28.71	31.39	37.21	44.41
29	13.56	15.13	15.93	18.22	19.49	21.04	23.83	27.05	29.85	32.61	38.63	46.07
30	14.25	15.86	16.68	19.03	20.34	21.93	24.80	28.11	30.99	33.84	40.05	47.73
31	14.94	16.60	17.44	19.85	21.19	22.83	25.77	29.17	32.14	35.07	41.46	49.39
32	15.63	17.34	18.20	20.68	22.05	23.72	26.74	30.24	33.28	36.29	42.88	51.05
33	16.33	18.08	18.97	21.50	22.91	24.63	27.72	31.30	34.43	37.52	44.30	52.71
34	17.04	18.83	19.74	22.33	23.77	25.53	28.70	32.37	35.57	38.75	45.71	54.38
35	17.75	19.59	20.52	23.17	24.64	26.43	29.68	33.43	36.72	39.98	47.14	56.04
36	18.47	20.35	21.29	24.01	25.51	27.34	30.66	34.50	37.87	41.21	48.55	57.70
37	19.19	21.11	22.08	24.85	26.38	28.25	31.64	35.57	39.02	42.45	49.98	59.36
38	19.91	21.87	22.86	25.69	27.25	29.17	32.62	36.64	40.17	43.68	51.39	61.03
39	20.64	22.64	23.65	26.53	28.13	30.08	33.61	37.71	41.32	44.91	52.82	62.69
40	21.37	23.41	24.44	27.38	29.01	31.00	34.60	38.79	42.47	46.14	54.23	64.35
41	22.11	24.19	25.24	28.23	29.89	31.91	35.58	39.86	43.63	47.38	55.66	66.02
42	22.84	24.97	26.04	29.08	30.77	32.84	36.57	40.93	44.78	48.61	57.08	67.68
43	23.59	25.75	26.84	29.94	31.65	33.76	37.56	42.01	45.93	49.85	58.50	69.34
44	24.33	26.53	27.64	30.80	32.54	34.68	38.56	43.09	47.09	51.09	59.92	71.00
45	25.08	27.32	28.45	31.65	33.43	35.61	39.55	44.16	48.24	52.32	61.34	72.67
46	25.83	28.11	29.25	32.52	34.32	36.53	40.54	45.24	49.40	53.56	62.77	74.33
47	26.59	28.90	30.06	33.38	35.21	37.46	41.54	46.32	50.55	54.79	64.19	76.00
48	27.34	29.70	30.88	34.24	36.11	38.39	42.54	47.40	51.71	56.03	65.61	77.66
49	28.10	30.49	31.69	35.11	37.00	39.32	43.53	48.48	52.87	57.27	67.04	79.32
50	28.87	31.29	32.51	35.98	37.90	40.25	44.53	49.56	54.03	58.51	68.46	80.98

S/B	0.01%	0.05%	0.1%	0.5%	1%	2%	5%	10%	15%	20%	30%	40%
51	29.63	32.09	33.33	36.85	38.80	41.19	45.53	50.64	55.18	59.75	69.88	82.65
52	30.40	32.90	34.15	37.72	39.70	42.12	46.53	51.72	56.34	60.98	71.30	84.31
53	31.17	33.70	34.98	38.60	40.60	43.06	47.53	52.80	57.50	62.22	72.73	85.98
54	31.94	34.51	35.80	39.47	41.50	44.00	48.54	53.89	58.66	63.46	74.15	87.64
55	32.71	35.32	36.63	40.35	42.41	44.93	49.54	54.97	59.82	64.70	75.57	89.30
56	33.49	36.13	37.46	41.23	43.31	45.88	50.54	56.06	60.98	65.94	77.00	90.97
57	34.27	36.95	38.29	42.11	44.22	46.81	51.55	57.14	62.14	67.18	78.42	92.63
58	35.05	37.76	39.12	42.99	45.13	47.75	52.55	58.23	63.30	68.42	79.84	94.30
59	35.84	38.58	39.96	43.87	46.04	48.70	53.56	59.31	64.46	69.66	81.27	95.96
60	36.62	39.40	40.79	44.75	46.95	49.64	54.56	60.40	65.63	70.90	82.70	97.63
61	37.41	40.22	41.63	45.64	47.86	50.59	55.57	61.48	66.79	72.14	84.12	99.30
62	38.20	41.04	42.47	46.53	48.77	51.53	56.58	62.57	67.95	73.38	85.55	101.0
63	38.99	41.87	43.31	47.41	49.69	52.48	57.59	63.66	69.11	74.63	86.97	102.6
64	39.78	42.70	44.15	48.30	50.60	53.43	58.60	64.75	70.27	75.86	88.39	104.3
65	40.58	43.52	45.00	49.19	51.52	54.38	59.61	65.84	71.44	77.11	89.82	106.0
66	41.38	44.35	45.84	50.09	52.43	55.32	60.62	66.93	72.60	78.34	91.24	107.6
67	42.17	45.18	46.69	50.98	53.35	56.27	61.63	68.02	73.77	79.59	92.67	109.3
68	42.97	46.01	47.54	51.87	54.27	57.22	62.64	69.11	74.93	80.83	94.09	110.9
69	43.77	46.85	48.39	52.77	55.19	58.18	63.65	70.20	76.09	82.07	95.52	112.6
70	44.57	47.68	49.24	53.66	56.11	59.13	64.66	71.29	77.26	83.31	96.95	114.3
71	45.38	48.52	50.09	54.55	57.03	60.08	65.68	72.38	78.42	84.55	98.37	115.9
72	46.19	49.36	50.94	55.45	57.95	61.04	66.69	73.46	79.59	85.80	99.80	117.6
73	46.99	50.19	51.80	56.35	58.88	61.99	67.71	74.55	80.75	87.04	101.2	119.3
74	47.80	51.04	52.65	57.25	59.80	62.94	68.72	75.65	81.91	88.29	102.6	120.9
75	48.61	51.88	53.51	58.15	60.73	63.90	69.73	76.74	83.08	89.53	104.1	122.6
76	49.43	52.72	54.37	59.05	61.65	64.86	70.75	77.83	84.24	90.77	105.5	124.3
77	50.24	53.56	55.23	59.95	62.58	65.81	71.77	78.92	85.41	92.02	106.9	125.9
78	51.05	54.41	56.09	60.86	63.50	66.77	72.79	80.02	86.58	93.26	108.4	127.6
79	51.87	55.25	56.95	61.76	64.43	67.73	73.80	81.11	87.74	94.50	109.8	129.3
80	52.68	56.10	57.81	62.66	65.36	68.69	74.82	82.20	88.91	95.74	111.2	130.9
81	53.50	56.95	58.67	63.57	66.29	69.64	75.84	83.29	90.07	96.99	112.6	132.6
82	54.32	57.80	59.54	64.48	67.22	70.61	76.86	84.38	91.24	98.23	114.1	134.3
83	55.14	58.65	60.40	65.38	68.15	71.57	77.87	85.48	92.41	99.48	115.5	135.9
84	55.96	59.50	61.27	66.29	69.08	72.53	78.89	86.57	93.57	100.7	116.9	137.6
85	56.79	60.35	62.13	67.20	70.01	73.49	79.91	87.67	94.74	102.0	118.3	139.3
86	57.61	61.20	63.00	68.11	70.95	74.45	80.93	88.77	95.91	103.2	119.8	140.9
87	58.44	62.06	63.87	69.02	71.88	75.41	81.95	89.86	97.07	104.5	121.2	142.6
88	59.27	62.91	64.74	69.93	72.81	76.38	82.97	90.95	98.24	105.7	122.6	144.3
89	60.09	63.77	65.61	70.84	73.75	77.34	83.99	92.05	99.41	106.9	124.0	145.9
90	60.92	64.63	66.48	71.75	74.68	78.30	85.01	93.14	100.6	108.2	125.5	147.6
91	61.75	65.48	67.36	72.66	75.62	79.27	86.03	94.23	101.7	109.4	126.9	149.3
92	62.58	66.34	68.23	73.58	76.55	80.23	87.05	95.34	102.9	110.7	128.3	150.9
93	63.41	67.20	69.10	74.49	77.49	81.20	88.08	96.43	104.1	111.9	129.7	152.6
94	64.25	68.07	69.98	75.41	78.43	82.16	89.09	97.52	105.3	113.2	131.2	154.3
95	65.08	68.93	70.85	76.32	79.37	83.13	90.12	98.63	106.4	114.4	132.6	155.9
96	65.91	69.79	71.73	77.24	80.30	84.09	91.14	99.72	107.6	115.7	134.0	157.6
97	66.75	70.65	72.61	78.16	81.24	85.06	92.16	100.8	108.8	116.9	135.5	159.3
98	67.59	71.52	73.48	79.07	82.18	86.03	93.19	101.9	109.9	118.1	136.9	160.9
99	68.43	72.38	74.36	79.98	83.12	87.00	94.21	103.0	111.1	119.4	138.3	162.6
100	69.26	73.25	75.24	80.91	84.06	87.97	95.23	104.1	112.3	120.6	139.7	164.3

*End of **Erlang B** Traffic Table*